

Data Cleaning and Preprocessing – Netflix Movies and TV Shows Dataset

Project Description

This project focuses on cleaning and preprocessing the Netflix Movies and TV Shows dataset as part of a Data Analyst Internship Task. The primary objective was to transform raw data into a structured, consistent, and analysis-ready format by addressing common data quality issues.

Objective

The goal of this project is to:

- Identify and handle missing values.
- Remove duplicate records.
- Standardize text and categorical data.
- Convert and format date fields.
- Validate and correct data types.
- Perform basic outlier analysis.
- Generate a cleaned dataset suitable for further analysis.

Dataset Information

Dataset Name: Netflix Movies and TV Shows Dataset

Source: Kaggle

Format: CSV

Technologies Used

- Python 3
- Pandas
- Visual Studio Code

Data Cleaning Process

1. Missing Value Handling

- Identified missing values using `isnull().sum()`.
- Replaced missing categorical values with "Unknown".
- Applied forward-fill technique for missing date values.

2. Duplicate Removal

- Identified duplicate records using `duplicated()`.
- Removed duplicate rows using `drop_duplicates()`.

3. Data Standardization

- Removed extra spaces.
- Standardized categorical values using title case formatting.
- Improved consistency across text-based columns.

4. Date Formatting

- Converted date fields into datetime format.
- Standardized date format to DD-MM-YYYY.

5. Column Standardization

- Converted column names to lowercase.
- Replaced spaces with underscores.

6. Data Type Validation

- Reviewed and validated column data types.
- Ensured numerical fields were stored in appropriate formats.

7. Outlier Detection

- Applied the Interquartile Range (IQR) method.
- Identified potential outliers within numerical fields.

Project Structure

Data-Cleaning-Task-1/

```
|  
|— netflix_titles.csv  
|— cleaned_netflix_titles.csv  
|— task1.py  
|— README.md  
|— summary_report.txt  
|— screenshots/
```

Files Included

File	Description
netflix_titles.csv	Original dataset
cleaned_netflix_titles.csv	Cleaned dataset

File	Description
task1.py	Python cleaning script
README.md	Project documentation
summary_report.txt	Task summary report
screenshots	Execution screenshots

Key Learning Outcomes

- Data preprocessing using Pandas
- Missing value treatment techniques
- Duplicate record management
- Data standardization practices
- Date conversion and formatting
- Data quality validation
- Outlier detection using IQR

Result

The dataset was successfully cleaned and transformed into a structured, analysis-ready format. The final cleaned dataset can be used for exploratory data analysis, dashboard creation, reporting, and predictive analytics projects.

Author

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Data Analyst Internship Task Submission